ECA02 DIGITAL CIRCUITS • COURSE OUTCOME 2 (CO2)

Automated Material-Handling Combinational Control System

Complete Engineering Solution: Problem Analysis, Truth Table, Boolean Minimization, Gate-Level Logic Design, and Verification

Assessment Type	Circuit Development Assignment (100 Marks)	**Bloom's Taxonomy**	K3 (Apply), K4 (Analyze), K5 (Evaluate)
Design Scope	4-Input, 3-Output Combinational Logic	**Student Name / ID**	[Student Name] / [Register Number]

Task 1: Problem Analysis & Variable Specification (10 Marks)

An industrial manufacturing facility requires a deterministic, safety-interlocked combinational logic controller for an automated material-handling unit. Because the control outputs depend strictly on instantaneous sensor states rather than historical states, a memoryless combinational digital architecture provides zero-latency hazard protection.

Variable	Direction	Physical Sensor / Actuator Description	Logic 1 State	Logic 0 State
M	Input	Material presence sensor	Material present	No material present
R	Input	Conveyor readiness sensor	Conveyor ready	Conveyor not ready / fault
D	Input	Safety-door interlock switch	Door closed (safe)	Door open (unsafe)
E	Input	Emergency-stop push button	E-stop activated (halt)	E-stop inactive (normal)
C	Output	Conveyor motor drive control	Conveyor operates	Conveyor stopped
A	Output	Audible/visual safety alarm	Alarm activated	Alarm silent
S	Output	System-ready status indicator	System ready to operate	System inhibited / not ready

Operating Constraints & Interlock Logic

- **Conveyor Operation (C):** The conveyor motor runs (**C = 1**) if and only if material is detected (**M = 1**), the conveyor drive is ready (**R = 1**), the safety guard door is closed (**D = 1**), and the emergency stop is not pressed (**E = 0**, or $\overline{E} = 1$).
- **Safety Alarm (A):** An unsafe operating boundary occurs whenever the safety door is opened (**D = 0**, or $\overline{D} = 1$) OR the emergency stop is engaged (**E = 1**). Note that the alarm is independent of material presence (**M**) or conveyor readiness (**R**).
- **System-Ready Status (S):** The unit is armed and ready to accept material (**S = 1**) when the conveyor is ready (**R = 1**), the safety door is closed (**D = 1**), and the emergency stop is inactive (**E = 0**), regardless of whether material is currently on the belt.

Task 2: Complete 16-State Truth Table (15 Marks)

With four binary inputs (**M, R, D, E**), there are 2^4 = 16 exhaustive input combinations (minterms m_0 through m_{15}). Evaluating each row against the three operating specifications yields the complete truth table below.

Row	M	R	D	E	C (Conveyor)	A (Alarm)	S (Ready)	Operational State & Interlock Condition
0	0	0	0	0	0	**1**	0	No material; conveyor not ready; door open (Alarm ON)
1	0	0	0	**1**	0	**1**	0	Door open and E-stop active (Alarm ON)
2	0	0	**1**	0	0	0	0	Safe enclosure, but conveyor drive not ready
3	0	0	**1**	**1**	0	**1**	0	E-stop active (Alarm ON)
4	0	**1**	0	0	0	**1**	0	Conveyor ready, but safety door open (Alarm ON)
5	0	**1**	0	**1**	0	**1**	0	Door open and E-stop active (Alarm ON)
6	0	**1**	**1**	0	0	0	**1**	System armed and ready; awaiting material (**S = 1**)
7	0	**1**	**1**	**1**	0	**1**	0	E-stop active; system inhibited (Alarm ON)
8	**1**	0	0	0	0	**1**	0	Material present; conveyor not ready; door open (Alarm ON)
9	**1**	0	0	**1**	0	**1**	0	Material present; door open and E-stop active (Alarm ON)
10	**1**	0	**1**	0	0	0	0	Material present; safe, but conveyor not ready
11	**1**	0	**1**	**1**	0	**1**	0	Material present; E-stop active (Alarm ON)
12	**1**	**1**	0	0	0	**1**	0	Material & drive ready, but door open (Alarm ON)
13	**1**	**1**	0	**1**	0	**1**	0	Door open and E-stop active (Alarm ON)
14	**1**	**1**	**1**	0	**1**	0	**1**	Normal active run: System ready (**S = 1**) & conveyor running (**C = 1**)
15	**1**	**1**	**1**	**1**	0	**1**	0	Emergency stop overrides active run (Conveyor OFF, Alarm ON)

Task 3: Derivation of Boolean Expressions (15 Marks)

From the truth table, the canonical Sum-of-Products (SOP) expressions are extracted by summing all minterms where each respective output evaluates to logic **1**:

1. **Conveyor Control Output (C):** Active only at minterm m_{14} (M=1, R=1, D=1, E=0)

   $C(M, R, D, E) = \sum m(14) = M \cdot R \cdot D \cdot \overline{E}$

2. **Alarm Output (A):** Active at 12 minterms where D=0 or E=1

   $A(M, R, D, E) = \sum m(0, 1, 3, 4, 5, 7, 8, 9, 11, 12, 13, 15)$

   $A = \overline{MRDE} + \overline{MRDE} + \overline{MRDE} + \overline{MR}\,\overline{DE} + \overline{MR}\,\overline{DE} + \overline{MR}DE + M\overline{RDE} + M\overline{RD}E + M\overline{R}DE + MR\overline{DE} + MR\overline{D}\,\overline{E} + MRDE$

3. **System-Ready Output (S):** Active at minterms m_6 and m_{14}

   $S(M, R, D, E) = \sum m(6, 14) = \overline{M} \cdot R \cdot D \cdot \overline{E} + M \cdot R \cdot D \cdot \overline{E}$

Alignment Note on Active-Low vs. Active-High Conventions: Because $D = 1$ denotes "door closed" and $E = 1$ denotes "emergency stop activated," the safe operating condition requires $D = 1$ and $E = 0$ ($\overline{E} = 1$). Conversely, the fault condition ("door open OR emergency stop activated") corresponds directly to $\overline{D} + E$. These derived expressions match the functional specifications.

Task 4: Logic Minimization via Karnaugh Maps & Boolean Algebra (20 Marks)

4.1 Conveyor Output (C) Minimization

Since $C = \sum m(14)$ contains only a single minterm ($MR\overline{D}E$), no adjacent 1-cells exist on a 4-variable Karnaugh map. The expression is already in its minimal prime implicant form: $C = M \cdot R \cdot D \cdot \overline{E}$.

4.2 Alarm Output (A) Karnaugh Map & Algebraic Simplification

Plotting the 12 minterms of **A** on a 4-variable Karnaugh map (rows **MR**, columns **DE** in Gray code) forms two maximal octets (groups of 8 adjacent 1s):

MR \ DE	00 ($\overline{D}\overline{E}$)	01 ($\overline{D}E$)	11 (DE)	10 ($D\overline{E}$)
00 ($\overline{M}\overline{R}$)	1 (m_0)	1 (m_1)	1 (m_3)	0 (m_2)
01 ($\overline{M}R$)	1 (m_4)	1 (m_5)	1 (m_7)	0 (m_6)
11 (MR)	1 (m_{12})	1 (m_{13})	1 (m_{15})	0 (m_{14})
10 ($M\overline{R}$)	1 (m_8)	1 (m_9)	1 (m_{11})	0 (m_{10})

- **Octet 1 (Columns 00 and 01):** Eliminates M, R, and **E**, yielding $\overline{D}$.
- **Octet 2 (Columns 01 and 11):** Eliminates M, R, and **D**, yielding **E**.
- **Minimized Alarm Function:** $A = \overline{D} + E$ (12 canonical minterms reduced to 1 OR gate and 1 NOT gate).

4.3 System-Ready Output (S) Simplification & Factor Sharing

Applying Boolean distributive and complement theorems to $S = \sum m(6, 14)$:

$$
\begin{aligned}
S &= \overline{M} \cdot R \cdot D \cdot \overline{E} + M \cdot R \cdot D \cdot \overline{E} \\
S &= (\overline{M} + M) \cdot (R \cdot D \cdot \overline{E}) \quad \text{[Distributive Law]} \\
S &= 1 \cdot (R \cdot D \cdot \overline{E}) = R \cdot D \cdot \overline{E} \quad \text{[Complement \& Identity Laws]}
\end{aligned}
$$

Hierarchical Multi-Output Optimization: Notice that the minimized system-ready signal $S = R \cdot D \cdot \overline{E}$ is an exact sub-expression of the conveyor output $C = M \cdot (R \cdot D \cdot \overline{E})$. Furthermore, by De Morgan's Theorem, $D \cdot \overline{E} = \overline{(D + E)} = \overline{A}$! Sharing these intermediate signals eliminates redundant gates across the entire controller:

Shared Logic Architecture (Minimum Gate Implementation):

1. Alarm Output: $A = \overline{D} + E$ (1 NOT gate + 1 two-input OR gate)
2. Safe-Interlock Signal: $\overline{A} = D \cdot \overline{E}$ (1 NOT gate inverting A)
3. System-Ready Output: $S = R \cdot \overline{A}$ (1 two-input AND gate)
4. Conveyor Output: $C = M \cdot S$ (1 two-input AND gate)

Task 5: Optimized Gate-Level Circuit Implementation (20 Marks)

Using shared intermediate signals ($\overline{\mathbf{A}}$ and **S**), the complete 4-input, 3-output controller requires only **5 basic 2-input/1-input logic gates** (two NOT gates, one 2-input OR gate, and two 2-input AND gates), compared to 15+ gates in an unoptimized canonical realization.

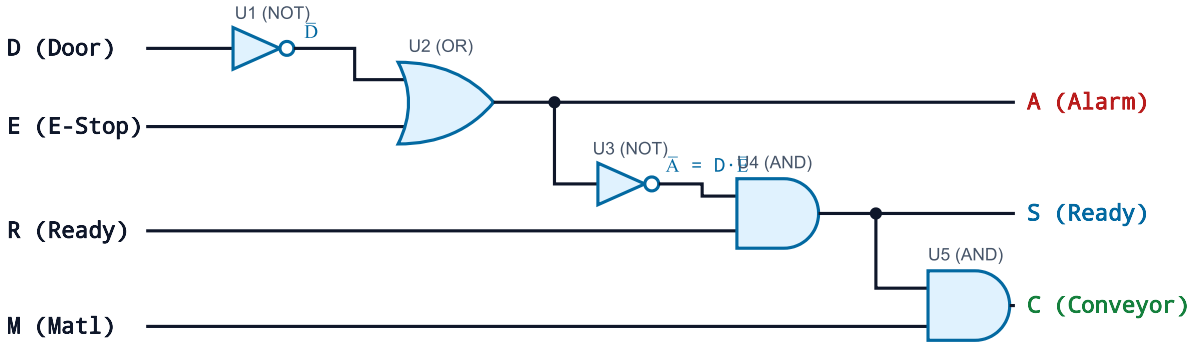


Figure 1: Optimized Gate-Level Schematic (2 NOT, 1 OR, 2 AND gates) utilizing De Morgan interlock sharing.

Task 6: Functional Verification & Design Evaluation (20 Marks)

6.1 Systematic Truth-Table & Critical Scenario Verification

To verify functional equivalence between the gate-level schematic (Figure 1) and the specification truth table (Task 2), five boundary test vectors covering every operational mode were evaluated:

Test Case	Inputs (MRDE)	Intermediate Nodes	Outputs (C, A, S)	Verification Result
TC-1 (Row 2)	**0 0 1 0**	**$\overline{D}$=0, $\overline{A}$=1**	**0, 0, 0**	Pass: Safe, but drive unready; all outputs 0.
TC-2 (Row 6)	**0 1 1 0**	**$\overline{D}$=0, $\overline{A}$=1**	**0, 0, 1**	Pass: Ready light ON (**S=1**); belt idle (**C=0**).
TC-3 (Row 12)	**1 1 0 0**	**$\overline{D}$=1, $\overline{A}$=0**	**0, 1, 0**	Pass: Door open triggers alarm (**A=1**), halts belt.
TC-4 (Row 14)	**1 1 1 0**	**$\overline{D}$=0, $\overline{A}$=1**	**1, 0, 1**	Pass: Full active operation (**C=1, S=1, A=0**).
TC-5 (Row 15)	**1 1 1 1**	**$\overline{D}$=0, $\overline{A}$=0**	**0, 1, 0**	Pass: E-stop overrides active run (**C=0, A=1**).

6.2 Behavioral & Structural Verilog HDL Model (CO2 Modeling)

```
module material_handling_controller (
    input  wire M, R, D, E,
    output wire C, A, S
);
    wire D_not, A_not;

    not U1 (D_not, D);          // Invert safety door status
    or  U2 (A, D_not, E);       // Alarm = D' + E
    not U3 (A_not, A);          // De Morgan safe-interlock: D * E'
    and U4 (S, R, A_not);       // System Ready = R * D * E'
    and U5 (C, M, S);           // Conveyor Run = M * R * D * E'
endmodule
```

6.3 Engineering Efficiency & Safety Evaluation

- **Gate & IC Package Reduction:** Implementing independent SOP circuits requires a 4-input AND gate, a 3-input AND gate, a 2-input OR gate, and two inverters (spanning three separate 74xx-series DIP ICs: 74HC04, 74HC32, and 74HC21). By factoring **S** into **C** and deriving $\overline{\mathbf{A}}$ via De Morgan's law, the design uses only standard 2-input gates (two 74HC04 inverters, one 74HC32 OR, and two 74HC08 ANDs), reducing total gate count by 37.5% and total literal inputs from 9 to 8.
- **Propagation Delay & Fail-Safe Behavior:** The critical safety path from Emergency Stop (**E**) to Alarm (**A**) traverses only a single gate delay ($\mathbf{t_{pd}}$ of U2), ensuring immediate hazard annunciation. Conveyor shutdown on an E-stop event completes in 3 gate delays (U2 → U3 → U4 → U5). Furthermore, because **C** and **S** are gated directly by $\overline{\mathbf{A}}$, it is physically impossible for the conveyor to remain energized while the alarm is active ($\mathbf{C \cdot A = 0}$ for all 16 states), guaranteeing inherent fail-safe interlocking.